

Relativity Toolkit, Part 1:

Cloud Load Strategy

```
In[*]:= (*1. Clean the kernel to ensure we aren't seeing cached definitions*)

FrontEndTokenExecute["DeleteGeneratedCells"];
Quit[]

In[1]:= (*2. LOAD THE CODE*)
branch = "main";
dlRunFile[nym_String] :=
  Module[{base, rules, tests, viz, bust},
    base = "https://raw.githubusercontent.com/rebcabin/RelativityToolkit/" <>
      branch <> "/";
    bust = "?t=" <> ToString[Round[AbsoluteTime[]]];
    (*Force fresh download*)
    Get[base <> nym <> bust];

In[3]:= Module[{base, rules, tests, viz, bust},
  base =
    "https://raw.githubusercontent.com/rebcabin/RelativityToolkit/" <> branch <> "/";
  bust = "?t=" <> ToString[Round[AbsoluteTime[]]];
  (*Force fresh download*)
  (*Define URLs*)
  rules = base <> "RelativityToolkit.wl" <> bust;
  tests = base <> "RelativityToolkit_RegressionTests.wl" <> bust;
  viz = base <> "RelativityToolkit_VisualShowcase.wl" <> bust;
  (*Load them*)
  Get[rules];
  Get[tests];
  Get[viz](*no semicolon on purpose to see gallery*)

=====

RUNNING REGRESSION SUITE v1.6.1

=====

--- SECTION 1: TYPE CHECKING (VALENCE) ---

[PASS] Vector  $x^\mu \rightarrow \{\{\mu\}, \{\}\}$ 
[PASS] Covector  $p_\nu \rightarrow \{\{\}, \{\nu\}\}$ 
```

[PASS] Mixed Tensor $T^\mu_\nu \rightarrow \{\{\mu\}, \{\nu\}\}$
 [PASS] Mixed Tensor $T_{\mu\nu} \rightarrow \{\{\}, \{\mu, \nu\}\}$
 [PASS] Mixed Tensor $T^{\mu\nu} \rightarrow \{\{\mu, \nu\}, \{\}\}$
 [PASS] Mixed Tensor $T_\mu{}^\nu \rightarrow \{\{\nu\}, \{\mu\}\}$
 [PASS] Scalar Contraction $x^\mu p_\mu \rightarrow \{\{\}, \{\}\}$
 [PASS] Scalar Contraction $p_\mu x^\mu \rightarrow \{\{\}, \{\}\}$
 [PASS] Tensor-Vector $T^\mu_\nu x^\nu \rightarrow x^\mu \rightarrow \{\{\mu\}, \{\}\}$
 [PASS] Tensor-Vector $T^\mu_\nu x^\mu \rightarrow x_\nu \rightarrow \{\{\}, \{\nu\}\}$
 [PASS] Power of Scalar $\rightarrow \{\{\}, \{\}\}$
 [PASS] Scalar Multiplication $\rightarrow \{\{\mu\}, \{\}\}$
 [PASS] Gradient $d(f)/dx^\alpha \rightarrow \text{Down}[\alpha] \rightarrow \{\{\}, \{\alpha\}\}$

Valence Mismatch

» expect error message above $\{\{\}, \{\}\}$

[PASS] Mismatch Handled Gracefully $\rightarrow \{\{\}, \{\}\}$

--- SECTION 2: ALGEBRAIC CANONICALIZATION ---

[PASS] Dummy Renaming $(A^\mu B_\mu == A^\nu B_\nu) \rightarrow (p_{i1}) (x^{i1})$
 [PASS] Commutativity $(A^\mu B_\mu == B_\nu A^\nu) \rightarrow (p_{i1}) (x^{i1})$
 [PASS] Index Coalescing $(A^\mu B_\mu + A^\nu B_\nu == 2 A^\rho B_\rho) \rightarrow 2 (p_{i1}) (x^{i1})$
 [PASS] Double Dummy Pairs $(P^\mu Q_\mu S^\nu T_\nu) \rightarrow (P^{i1}) (Q_{i1}) (S^{i2}) (T_{i2})$
 [PASS] Free Indices Preserved $\rightarrow \{(A^\alpha) (B^\beta), (A^{i1}) (B^{i1})\}$

--- SECTION 3: PHYSICS, METRIC & TRANSFORMATIONS ---

[PASS] Lowering Index $(g_{\mu\nu} A^\nu \rightarrow A_\mu) \rightarrow A_\mu$
 [PASS] Raising Index $(g^{\mu\nu} p_\nu \rightarrow p^\mu) \rightarrow p^\mu$
 [PASS] Inverse Metric $(g^{\mu\alpha} g_{\alpha\nu} \rightarrow \delta^\mu_\nu) \rightarrow \delta^\mu_\nu$
 [PASS] Alpha-Conversion: distinct indices $\rightarrow \{\mu11, \mu12\}$
 [PASS] Triple Metric Sandwich $(g.g.g \rightarrow g) \rightarrow g_{\mu\nu}$
 [PASS] Delta-Gamma Contraction $\rightarrow \Gamma^S_{ab}$

--- SECTION 4: METRIC EQUIVALENCE $(A^\mu B_\mu == A_\nu B^\nu)$ ---

[PASS] Structural Distinction $A^\mu B_\mu := A_\nu B^\nu$ initially $\rightarrow \{(A^{i1}) (B_{i1}), (A_{i1}) (B^{i1})\}$

$\mathcal{UD}$ will rearrange terms

[PASS] Metric Equivalence $(A_\nu B^\nu \text{ reduces to } A^\mu B_\mu) \rightarrow (A^{i1}) (B_{i1})$

--- SECTION 5: SECOND DERIVATIVE Box Structure ---

[PASS] Found FractionBox + ^2 $\rightarrow \text{True}$

--- SECTION 6: CALCULUS ENGINE ---

[PASS] Leibniz Product Rule $\rightarrow g f_{,\mu} + f g_{,\mu}$
 [PASS] Covariant Derivative Expansion $\rightarrow \text{True}$
 [PASS] Differentiation Linearity $\rightarrow \text{True}$
 [PASS] CD of Scalar reduces to Partial $\rightarrow \text{phi}_{10458,\mu}$
 [PASS] Scalar CD contains no Connection Coefficients $\rightarrow \text{True}$
 [PASS] Nested CD Alpha-Conversion (Unique Indices Generated) $\rightarrow \text{True}$

--- SECTION 7: QUOTIENT THEOREM (Extraction) ---

[PASS] Quotient Extraction (Simple) $\rightarrow T_{\text{S}}$
 [PASS] Quotient Extraction (Distributed Dummies) $\rightarrow R_{\text{S}} + S_{\text{S}}$
 [PASS] Quotient Extraction (Zero Case) $\rightarrow 0$

--- SECTION 8: TENSOR FORM ROBUSTNESS ---

[PASS] TensorForm maps arbitrary formals (Q, Z, I99) to Greek $\rightarrow \text{True}$

--- SECTION 9: COMMA NOTATION DISPLAY LOGIC ---

[PASS] Atomic Tensor $\rightarrow A^{\mu}_{,\nu}$
 [PASS] Sealed Product (Gamma Glue) $\rightarrow ((A^{\mu}) \Gamma^a_{bc})_{,\nu}$
 [PASS] Sealed Sum $\rightarrow (A+B)_{,\nu}$
 [PASS] Sealed Power $\rightarrow (A^2)_{,\nu}$
 [PASS] Arbitrary Function (Sin) $\rightarrow \text{Sin}[\text{theta}]_{,\nu}$
 [PASS] Nested Math $(A \star (B+C)) \rightarrow (A (B+C))_{,\nu}$

 REGRESSION SUITE COMPLETE

PASSED: 44

FAILED: 0

STATUS: GREEN (Ready for Publication)

=====

VISUAL SHOWCASE: FORMAL DIFFERENTIAL GEOMETRY

v1.6.1

=====

1. The Tensor Zoo (Standard Objects)

These objects carry valence metadata hidden inside their structure.

Object Name	Wolfram Input	Textbook Output
Contravariant Vector	<code>x[U[μ]]</code>	x^μ
Covariant Covector	<code>p[D[v]]</code>	p_ν
Mixed Tensor	<code>T[U[μ], D[v]]</code>	$T^\mu{}_\nu$
The Metric	<code>g[D[μ], D[ν]]</code>	$g_{\mu\nu}$
Inverse Metric	<code>g[U[μ], U[ν]]</code>	$g^{\mu\nu}$
Kronecker Delta	<code>δ[U[μ], D[ν]]</code>	δ^μ_ν
Jacobian (Partial)	<code>Partials[x[U[α]], x[U[β]]]</code>	$\frac{\partial x^\alpha}{\partial x^\beta}$

2. Metric Gymnastics (Raising & Lowering)

Watch the metric 'swallow' indices to change their variance.

Index Lowering
Input: $(A^\nu)(g_{\mu\nu})$
Result: A_μ
Index Raising
Input: $(g^{\mu\nu})(p_\nu)$
Result: p^μ
The Inverse Identity
Input: $(g_{\alpha\nu})(g^{\mu\alpha})$
Result: δ^μ_ν

3. Coordinate Transformations (Alpha-Conversion)

Note the automatically generated unique dummy index (e.g., $\mu\$...$).

Transforming a Vector
Input: $A^{a'}$
Result: $(A^{\mu2\theta}) \frac{\partial x^{\mu2\theta}}{\partial x^{a'}}$

4. Canonicalization & Pretty Printing

Simplifying dummy indices into standard Greek letters.

Raw Sum (Different Indices): $(A^\mu)(B_\mu) + (A^\nu)(B_\nu)$

Canonicalized (TensorForm): $2(A^\lambda)(B_\lambda)$

5. Physical Equivalence Proof

Demonstrating that $A^\mu B_\mu$ is equivalent to $A_\nu B^\nu$.

($\mathcal{U}\mathcal{D}$ will reorganize and canonicalize displays.)

Step	Expression	Input
1. Start	$(A_\nu)(B^\nu)$	Input: $A[\mathcal{D}[\nu]]B[\mathcal{U}[\nu]]$
2. Definition	$(A^\alpha)(B^\nu)(g_{\nu\alpha})$	$(g[\mathcal{D}[\nu],\mathcal{D}[\alpha]]A[\mathcal{U}[\alpha]])B[\mathcal{U}[\nu]]$
3. Commute?	$(A^\alpha)(B^\nu)(g_{\nu\alpha})$	$A[\mathcal{U}[\alpha]](g[\mathcal{D}[\nu],\mathcal{D}[\alpha]]B[\mathcal{U}[\nu]])$
4. Contract	$(A_\nu)(B^\nu)$	step2Term/. metricRules
5. Equality?	True	lhs:4 === rhs:1

=====

--- Gallery of Derivatives ---

Description	Rendered Output
Generic Mixed Partial	$\frac{\partial^2 f}{\partial x \partial y}$
Coordinate Hessian	$\frac{\partial^2 x^{\alpha'}}{\partial x^{\beta} \partial x^{\alpha}}$
Atomic Tensor	$A^{\mu}_{,\nu}$
Sealed Product (v.1.6.1 fix)	$\left((A^{\mu}) \Gamma^{\rho}_{\sigma\lambda} \right)_{,\nu}$
Sealed Sum (Linearity)	$(A^{\mu} + B^{\mu})_{,\nu}$
Sealed Power (Non-Linear)	$\left(\phi^2 \right)_{,\nu}$
Covariant Derivative (Abstract)	$A^{\mu}_{;\nu}$
Covariant Derivative (Expanded)	$A^{\mu}_{,\nu} + (A^{\lambda}) \Gamma^{\mu}_{\nu\lambda}$

Out[3]=

```
In[4]:= dlRunFile["covariant_derivative.wl"];
```

- » 1. Input $\partial_{\beta'}(A^{\alpha'}) \equiv A^{\alpha'}_{,\beta'}$: $\left(A^{\mu 23} \right) \frac{\partial x^{ap}}{\partial x^{\mu 23}}_{,bp}$
- » 2. Mechanically Expanded Derivative: $\left(A^{\mu 23} \right) \frac{\partial^2 x^{ap}}{\partial x^{\mu 23} \partial x^{bp}} + \left(A^{\mu 23}_{,bp} \right) \frac{\partial x^{ap}}{\partial x^{\mu 23}}$
- » 3. Peek at unique dummy indices $\left(A^{\mu 23} \right) \frac{\partial^2 x^{\alpha'}}{\partial x^{\beta'} \partial x^{\mu 23}} + \left(A^{\mu 23}_{,\beta'} \right) \frac{\partial x^{\alpha'}}{\partial x^{\mu 23}}$
- » 4. Expanded $\partial_{\beta'}(A^{\alpha'})$ in Readable Greek $\left(A^{\lambda} \right) \frac{\partial^2 x^{\alpha'}}{\partial x^{\beta'} \partial x^{\lambda}} + \left(A^{\lambda}_{,\beta'} \right) \frac{\partial x^{\alpha'}}{\partial x^{\lambda}}$

In[5]:= **dlRunFile["law_of_gamma.wl"];**

- » 1. $\mathcal{UD}$ says $A^{\alpha'}_{,\beta'} = \frac{\partial x^{ap}}{\partial x^{\mu\nu}} \frac{\partial x^{\nu u}}{\partial x^{bp}} \left(A^{\mu\nu}_{,\nu u} + \left(A^{\lambda 24} \right) \Gamma^{\mu\nu}_{\nu\lambda 24} \right)$
- » 2a. $A^{\mu'} = \left(A^{\mu 25} \right) \frac{\partial x^{ap}}{\partial x^{\mu 25}}$
- » 2b. Broken partial $A^{\alpha'}_{,\beta'} = \left(A^{\mu 25} \right) \frac{\partial^2 x^{ap}}{\partial x^{\mu 25} \partial x^{bp}} + \left(A^{\mu 25}_{,\mu 26} \right) \frac{\partial x^{ap}}{\partial x^{\mu 25}} \frac{\partial x^{\mu 26}}{\partial x^{bp}}$
- » 3a. Residual $(A^{\alpha'}_{,\beta'} - A^{\alpha'}_{,\beta'}) = - \left(A^{\mu 25} \right) \frac{\partial^2 x^{\alpha'}}{\partial x^{\beta'} \partial x^{\mu 25}} -$
 $\left(A^{\mu 25}_{,\mu 26} \right) \frac{\partial x^{\mu 26}}{\partial x^{\beta'}} \frac{\partial x^{\alpha'}}{\partial x^{\mu 25}} + \frac{\partial x^{\nu}}{\partial x^{\beta'}} \frac{\partial x^{\alpha'}}{\partial x^{\mu}} \left(A^{\mu}_{,\nu} + \left(A^{\lambda 24} \right) \Gamma^{\mu}_{\nu\lambda 24} \right)$

=== Γ TRANSFORMATION LAW (without quotient) ===

The term involving Γ' must satisfy:

- » 4. $A^{\nu'} \Gamma^{\alpha'}_{\beta' \nu'} = - \left(A^{\lambda} \right) \frac{\partial^2 x^{\alpha'}}{\partial x^{\beta'} \partial x^{\lambda}} + \left(A^{\lambda} \right) \frac{\partial x^{\kappa}}{\partial x^{\beta'}} \frac{\partial x^{\alpha'}}{\partial x^{\rho}} \Gamma^{\rho}_{\kappa \lambda}$

=== Γ TRANSFORMATION LAW ===

By a Quotient Theorem, stripping A to find Γ' :

- » 5. $\Gamma^{\alpha'}_{\beta' \nu'} = - \frac{\partial^2 x^{\alpha'}}{\partial x^{\beta'} \partial x^{\lambda}} \frac{\partial x^{\lambda}}{\partial x^{\nu'}} + \frac{\partial x^{\kappa}}{\partial x^{\beta'}} \frac{\partial x^{\lambda}}{\partial x^{\nu'}} \frac{\partial x^{\alpha'}}{\partial x^{\rho}} \Gamma^{\rho}_{\kappa \lambda}$

Not Tensorial!

In[6]:= **dlRunFile["connections_to_christoffels.wl"]**

=====

Christoffel Symbols for Gravitation from Metric via the $\mathcal{UD}$ Calculus

=====

1. PHYSICAL CONSTRAINTS -----

1a. Torsion-free geometry: $\Gamma^{\alpha}_{\beta\gamma} == \Gamma^{\alpha}_{\gamma\beta}$, [MTW] Box 10.2, page 250

- » Γ Symmetry: $\Gamma[u_, \mathcal{D}[b_], \mathcal{D}[c_]] \rightarrow \Gamma[u, \mathcal{D}[\text{Sort}[\{b, c\}][[1]]], \mathcal{D}[\text{Sort}[\{b, c\}][[2]]]]$

1b. Metric Symmetry: $g_{\alpha\beta} == g_{\beta\alpha}$, [MTW] Equation 8.24, page 210

(output manipulated as expected by $\mathcal{UD}$ display rules)

- » g Symmetry $g_{a_ b_} \rightarrow g_{a b}$

1c. Metric Compatibility Postulates $\nabla_{[\lambda} g_{\mu\nu]} = 0$ [MTW] Eqn 8.23 & Chapter 14.

$$\gg 0 = \nabla_{\lambda} g_{\mu\nu} = g_{\mu\nu,\lambda} - (g_{\kappa\nu}) \Gamma_{\lambda\mu}^{\kappa} - (g_{\kappa\mu}) \Gamma_{\lambda\nu}^{\kappa}$$

$$\gg 0 = \nabla_{\mu} g_{\lambda\kappa} = g_{\lambda\kappa,\mu} - (g_{\kappa\nu}) \Gamma_{\lambda\mu}^{\kappa} - (g_{\lambda\nu}) \Gamma_{\mu\kappa}^{\nu}$$

$$\gg 0 = \nabla_{\nu} g_{\lambda\mu} = g_{\lambda\mu,\nu} - (g_{\kappa\mu}) \Gamma_{\lambda\nu}^{\kappa} - (g_{\lambda\kappa}) \Gamma_{\mu\nu}^{\kappa}$$

2. CYCLIC PERMUTATION TRICK -----

Write 'Koszul' sum of ∇g permutations that isolates a Γ term.

[MTW] Equation 8.24b, page 210, in coordinate basis ($C_{abc}=0$)

The Linear Combination $0 = (\nabla_{\mu} g_{\nu\lambda} + \nabla_{\nu} g_{\lambda\mu} - \nabla_{\lambda} g_{\mu\nu}) ==$

$$\gg 0 = g_{\lambda\mu,\nu} + g_{\lambda\nu,\mu} - g_{\mu\nu,\lambda} + (g_{\lambda 27\nu}) \Gamma_{\lambda\mu}^{27} + (g_{\lambda 28\mu}) \Gamma_{\lambda\nu}^{28} -$$

$$(g_{\lambda\lambda 29}) \Gamma_{\mu\nu}^{29} - (g_{\lambda 30\nu}) \Gamma_{\lambda\mu}^{30} - (g_{\lambda 31\mu}) \Gamma_{\lambda\nu}^{31} - (g_{\lambda\lambda 32}) \Gamma_{\mu\nu}^{32}$$

$$\gg \Gamma \text{ part} = (g_{\lambda 27\nu}) \Gamma_{\lambda\mu}^{27} + (g_{\lambda 28\mu}) \Gamma_{\lambda\nu}^{28} - (g_{\lambda\lambda 29}) \Gamma_{\mu\nu}^{29} - (g_{\lambda 30\nu}) \Gamma_{\lambda\mu}^{30} - (g_{\lambda 31\mu}) \Gamma_{\lambda\nu}^{31} - (g_{\lambda\lambda 32}) \Gamma_{\mu\nu}^{32}$$

$$\gg \Gamma \text{ part (canonical)} = -2 (g_{\lambda\lambda j1}) \Gamma_{\mu\nu}^{j1}$$

$$\gg g \text{ partials } g_{\lambda\mu,\nu} + g_{\lambda\nu,\mu} - g_{\mu\nu,\lambda}$$

3. SOLVE FOR Γ -----

3a. Equation to solve:

$$\gg \text{Isolated Equation: } 2 (g_{\lambda\kappa}) \Gamma_{\mu\nu}^{\kappa} == g_{\lambda\mu,\nu} + g_{\lambda\nu,\mu} - g_{\mu\nu,\lambda}$$

3b. Find correct g^{-1} to multiply through.

$$\gg g \text{ found } g_{\lambda j1}$$

$$\gg \Gamma \text{ found } \Gamma_{\mu\nu}^{j1}$$

$$\gg g \text{ indices } \{\lambda, j1\}$$

$$\gg \Gamma \text{ up index } j1$$

$$\gg \text{target index } \lambda$$

$$\gg \frac{1}{2} g^{-1} = \frac{1}{2} (g^{\sigma\lambda})$$

(1/2 cancels factor 2 from Γ part (canonical) above)

3c. Multiply and Contract

$$\gg \delta\text{-}\Gamma \text{ contraction rule: } \Gamma[\mathcal{U}[r_-], a_-, b_-] \delta_{r_-}^{\varepsilon_-} \rightarrow \Gamma[\mathcal{U}[s], a, b]$$

(will be promoted into $\mathcal{UD}$ in the future)

$$\gg \text{LHS solved: } \Gamma_{\mu\nu}^{\sigma}$$

$$\gg \text{RHS solved: } \frac{1}{2} (g^{\sigma\lambda}) (g_{\lambda\mu,\nu} + g_{\lambda\nu,\mu} - g_{\mu\nu,\lambda})$$

=== CHRISTOFFEL SYMBOLS FOR GRAVITATION ===

aka The Fundamental Theorem of Riemannian Geometry

$$\text{Out[6]= } \Gamma_{\mu\nu}^{\sigma} == \frac{1}{2} (g^{\sigma\lambda}) (g_{\lambda\mu,\nu} + g_{\lambda\nu,\mu} - g_{\mu\nu,\lambda})$$

In[7]:= **dlRunFile["calculate_riemann.wl"]**

- » $\nabla_\rho \nabla_\nu A^\mu$: $\frac{\partial^2 A^\mu}{\partial x^\rho \partial x^\nu} + \left(A^{\lambda 33} \right) \Gamma_{\nu \lambda 33}^\mu{}_{,\rho} - \left(A^{\mu, \lambda 35} \right) \Gamma_{\nu \rho}^{\lambda 35} -$
 $\left(A^{\lambda 33} \right) \Gamma_{\nu \rho}^{\lambda 37} \Gamma_{\lambda 37 \lambda 33}^\mu + \left(A^{\lambda 34}{}_{,\nu} \right) \Gamma_{\rho \lambda 34}^\mu + \left(A^{\lambda 33} \right) \Gamma_{\nu \lambda 33}^{\lambda 36} \Gamma_{\rho \lambda 36}^\mu$
- » $\nabla_\nu \nabla_\rho A^\mu$: $\frac{\partial^2 A^\mu}{\partial x^\rho \partial x^\nu} + \left(A^{\lambda 38} \right) \Gamma_{\rho \lambda 38}^\mu{}_{,\nu} - \left(A^{\mu, \lambda 40} \right) \Gamma_{\rho \nu}^{\lambda 40} -$
 $\left(A^{\lambda 38} \right) \Gamma_{\rho \nu}^{\lambda 42} \Gamma_{\lambda 42 \lambda 38}^\mu + \left(A^{\lambda 39}{}_{,\rho} \right) \Gamma_{\nu \lambda 39}^\mu + \left(A^{\lambda 38} \right) \Gamma_{\rho \lambda 38}^{\lambda 41} \Gamma_{\nu \lambda 41}^\mu$
- » **raw commutator** $\left(\left(A^{\lambda 33} \right) \Gamma_{\nu \lambda 33}^\mu{}_{,\rho} - \left(A^{\lambda 38} \right) \Gamma_{\rho \lambda 38}^\mu{}_{,\nu} - \right.$
 $\left. \left(A^{\mu, \lambda 35} \right) \Gamma_{\nu \rho}^{\lambda 35} + \left(A^{\mu, \lambda 40} \right) \Gamma_{\rho \nu}^{\lambda 40} - \left(A^{\lambda 33} \right) \Gamma_{\nu \rho}^{\lambda 37} \Gamma_{\lambda 37 \lambda 33}^\mu + \left(A^{\lambda 38} \right) \Gamma_{\rho \nu}^{\lambda 42} \Gamma_{\lambda 42 \lambda 38}^\mu - \right.$
 $\left. \left(A^{\lambda 39}{}_{,\rho} \right) \Gamma_{\nu \lambda 39}^\mu - \left(A^{\lambda 38} \right) \Gamma_{\rho \lambda 38}^{\lambda 41} \Gamma_{\nu \lambda 41}^\mu + \left(A^{\lambda 34}{}_{,\nu} \right) \Gamma_{\rho \lambda 34}^\mu + \left(A^{\lambda 33} \right) \Gamma_{\nu \lambda 33}^{\lambda 36} \Gamma_{\rho \lambda 36}^\mu \right.$
- » **xpd commutator**: $\left(A^{\lambda 33} \right) \Gamma_{\nu \lambda 33}^\mu{}_{,\rho} - \left(A^{\lambda 38} \right) \Gamma_{\rho \lambda 38}^\mu{}_{,\nu} - \left(A^{\mu, \lambda 35} \right) \Gamma_{\nu \rho}^{\lambda 35} + \left(A^{\mu, \lambda 40} \right) \Gamma_{\rho \nu}^{\lambda 40} -$
 $\left(A^{\lambda 33} \right) \Gamma_{\nu \rho}^{\lambda 37} \Gamma_{\lambda 37 \lambda 33}^\mu + \left(A^{\lambda 38} \right) \Gamma_{\rho \nu}^{\lambda 42} \Gamma_{\lambda 42 \lambda 38}^\mu + \left(A^{\lambda 33}{}_{,\rho} \right) \Gamma_{\nu \lambda 33}^\mu - \left(A^{\lambda 39}{}_{,\rho} \right) \Gamma_{\nu \lambda 39}^\mu -$
 $\left(A^{\lambda 38} \right) \Gamma_{\rho \lambda 38}^{\lambda 41} \Gamma_{\nu \lambda 41}^\mu + \left(A^{\lambda 34}{}_{,\nu} \right) \Gamma_{\rho \lambda 34}^\mu + \left(A^{\lambda 33} \right) \Gamma_{\nu \lambda 33}^{\lambda 36} \Gamma_{\rho \lambda 36}^\mu - \left(A^{\lambda 38}{}_{,\nu} \right) \Gamma_{\rho \lambda 38}^\mu$
- » **Expanded – SealedBox diagnostic**: $\left(A^{\lambda 33} \right) \Gamma_{\nu \lambda 33}^\mu{}_{,\rho} - \left(\left(A^{\lambda 33} \right) \Gamma_{\nu \lambda 33}^\mu{}_{,\rho} - \right.$
 $\left. \left(A^{\lambda 38} \right) \Gamma_{\rho \lambda 38}^\mu{}_{,\nu} + \left(\left(A^{\lambda 38} \right) \Gamma_{\rho \lambda 38}^\mu{}_{,\nu} + \left(A^{\lambda 33}{}_{,\rho} \right) \Gamma_{\nu \lambda 33}^\mu - \left(A^{\lambda 38}{}_{,\nu} \right) \Gamma_{\rho \lambda 38}^\mu \right.$
- » **alg. xp + torsion elmin**: $\left(A^{\dot{1}1} \right) \Gamma_{\dot{1}1 \nu, \rho}^\mu - \left(A^{\dot{1}1} \right) \Gamma_{\dot{1}1 \rho, \nu}^\mu - \left(A^{\dot{1}1} \right) \Gamma_{\dot{1}1 \rho}^{\dot{1}2} \Gamma_{\dot{1}2 \nu}^\mu + \left(A^{\dot{1}1} \right) \Gamma_{\dot{1}1 \nu}^{\dot{1}2} \Gamma_{\dot{1}2 \rho}^\mu$
- » **test: has 2d derivs?** False
- » **raw Riemann** $\Gamma_{\dot{5} \nu, \rho}^\mu - \Gamma_{\dot{5} \rho, \nu}^\mu - \Gamma_{\dot{5} \rho}^{\dot{1}2} \Gamma_{\dot{1}2 \nu}^\mu + \Gamma_{\dot{5} \nu}^{\dot{1}2} \Gamma_{\dot{1}2 \rho}^\mu$

=== THE RIEMANN CURVATURE TENSOR ===

» $R^\mu_{\lambda \nu \rho} = \Gamma_{\lambda \nu, \rho}^\mu - \Gamma_{\lambda \rho, \nu}^\mu - \Gamma_{\lambda \rho}^\kappa \Gamma_{\kappa \nu}^\mu + \Gamma_{\lambda \nu}^\kappa \Gamma_{\kappa \rho}^\mu$

In[8]:= **dlRunFile["compiler_poc.wl"]**

» $g_{mn} \left\{ \left\{ -1 + \frac{2M}{r}, 0, 0, 0 \right\}, \left\{ 0, \frac{1}{1 - \frac{2M}{r}}, 0, 0 \right\}, \left\{ 0, 0, r^2, 0 \right\}, \left\{ 0, 0, 0, r^2 \sin[\theta]^2 \right\} \right\}$

» Γ from $\mathcal{UD}$

$$\begin{pmatrix} \begin{pmatrix} 0 \\ \frac{M}{r^2 - 2rM} \\ 0 \\ 0 \end{pmatrix} & \begin{pmatrix} \frac{M}{r^2 - 2rM} \\ 0 \\ 0 \\ 0 \end{pmatrix} & \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} & \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \\ \begin{pmatrix} \frac{(r-2M)M}{r^3} \\ 0 \\ 0 \\ 0 \end{pmatrix} & \begin{pmatrix} 0 \\ -\frac{M}{r^2 - 2rM} \\ 0 \\ 0 \end{pmatrix} & \begin{pmatrix} 0 \\ 0 \\ -r + 2M \\ 0 \end{pmatrix} & \begin{pmatrix} 0 \\ 0 \\ 0 \\ -(r-2M)\sin[\theta]^2 \end{pmatrix} \\ \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} & \begin{pmatrix} 0 \\ 0 \\ \frac{1}{r} \\ 0 \end{pmatrix} & \begin{pmatrix} 0 \\ \frac{1}{r} \\ 0 \\ 0 \end{pmatrix} & \begin{pmatrix} 0 \\ 0 \\ 0 \\ -\cos[\theta]\sin[\theta] \end{pmatrix} \\ \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} & \begin{pmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{r} \end{pmatrix} & \begin{pmatrix} 0 \\ 0 \\ 0 \\ \cot[\theta] \end{pmatrix} & \begin{pmatrix} 0 \\ \frac{1}{r} \\ \cot[\theta] \\ 0 \end{pmatrix} \end{pmatrix}$$

» Γ from WFR

$$\begin{pmatrix} \begin{pmatrix} 0 \\ \frac{M}{r^2 - 2rM} \\ 0 \\ 0 \end{pmatrix} & \begin{pmatrix} \frac{M}{r^2 - 2rM} \\ 0 \\ 0 \\ 0 \end{pmatrix} & \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} & \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \\ \begin{pmatrix} \frac{(r-2M)M}{r^3} \\ 0 \\ 0 \\ 0 \end{pmatrix} & \begin{pmatrix} 0 \\ -\frac{M}{r^2 - 2rM} \\ 0 \\ 0 \end{pmatrix} & \begin{pmatrix} 0 \\ 0 \\ -r + 2M \\ 0 \end{pmatrix} & \begin{pmatrix} 0 \\ 0 \\ 0 \\ -(r-2M)\sin[\theta]^2 \end{pmatrix} \\ \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} & \begin{pmatrix} 0 \\ 0 \\ \frac{1}{r} \\ 0 \end{pmatrix} & \begin{pmatrix} 0 \\ \frac{1}{r} \\ 0 \\ 0 \end{pmatrix} & \begin{pmatrix} 0 \\ 0 \\ 0 \\ -\cos[\theta]\sin[\theta] \end{pmatrix} \\ \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} & \begin{pmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{r} \end{pmatrix} & \begin{pmatrix} 0 \\ 0 \\ 0 \\ \cot[\theta] \end{pmatrix} & \begin{pmatrix} 0 \\ \frac{1}{r} \\ \cot[\theta] \\ 0 \end{pmatrix} \end{pmatrix}$$

» $\text{computed } \Gamma == \text{ground truth from WFR}$ True